Status	Finished
Started	Thursday, 29 October 2026, 10:55 PM
Completed	Thursday, 29 October 2026, 11:01 PM
Duration	5 mins 32 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Merging

b. Searching

c. Complexity

d. Sorting

The correct answer is:
Complexity

Question **2**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

a. Divide and conquer

b. Dynamic programming

c. Greedy method

d. Backtracking

The correct answer is:
Divide and conquer

Question **3**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

- a. Removing duplicates from a list
- b. Arranging data in a specific order
- c. Merging two datasets
- d. Finding a specific element in a list

The correct answer is: Arranging data in a specific order

Question **4**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

- a. It sorts data out of the box efficiently
- b. It only works with integer arrays
- c. It requires external libraries
- d. It is less efficient than custom sorting algorithms

The correct answer is: It sorts data out of the box efficiently

Question **5**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

- a. Insertion
- b. Selection
- c. Distribution
- d. Extraction

The correct answer is:
Insertion

Question **6**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

a. It does not require recursion
b. It is less efficient than bubble sort
c. It is based on the divide-and-conquer approach
d. It sorts data using a single pass

The correct answer is: It is based on the divide-and-conquer approach

Question **7**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

a. They are compared element by element
b. They are split again into smaller sublists
c. They are combined to form a single sorted list
d. They are discarded

The correct answer is: They are combined to form a single sorted list

Question **8**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

a. Heap sort
b. Merge sort
c. Selection sort
d. Quick sort

The correct answer is:
Merge sort

Question **9**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

- a. Arranging data in a particular format
- b. Inserting data into a list
- c. Searching for data in a list
- d. Deleting data from a list

The correct answer is: Arranging data in a particular format

Question **10**

Complete

Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

- a. Quick Sort
- b. Bubble Sort
- c. Linear Search
- d. Merge Sort

The correct answer is: Merge Sort

Question **11**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

- a. The parts are merged without sorting
- b. Each part is searched for a specific element
- c. The parts are ignored
- d. Each part is sorted independently

The correct answer is: Each part is sorted independently

Question **12**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

- a. It makes it easier to select items based on their relationship to the rest of the items
- b. It slows down the process
- c. It makes the data unsorted
- d. It complicates the selection of items

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **13**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

- a. A sorting algorithm is stable if it preserves the order of all keys
- b. A sorting algorithm is stable if it preserves the order of duplicate keys
- c. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys
- d. A sorting algorithm is stable if it preserves the order of non-duplicate keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **14**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 80,150,200,470,40,90,300,400,70,120
- b. 200,470,80,150,40,90,300,400,70,120
- c. 40,80,90,150,200,300,400,470,70,120
- d. 40,70,80,90,120,150,200,300,400,470

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **15**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

a. It repeatedly finds the minimum element and moves it to the sorted part of the list
b. It builds a sorted array one element at a time
c. It divides the list into two halves, sorts each half, and then merges them
d. It compares adjacent elements and swaps them if necessary

The correct answer is: It divides the list into two halves, sorts each half, and then merges them